

Name _____ Date _____ Period _____

Parent Functions I: $\sqrt{x}$ and $|x|$

Algebra II Honors | Unit 1 | Day 4 | TEK A2.2.A

Objective: I can graph and analyze the square root and absolute value parent functions, write absolute value as a piecewise rule, and explain why the two behave differently for negative inputs.

Vocabulary

Parent function — the simplest member of a family, with nothing added or multiplied.

Piecewise rule — one function defined by _____ on different parts of the domain.

Principal square root — $\sqrt{a}$ means the _____ root only, which is why $\sqrt{x}$ is a function.

Part 1 — Build Both Tables

Fill both. Note where outputs are undefined and where they repeat.

x	-4	-1	0	1	4	9
$f(x) = \sqrt{x}$						

x	-4	-1	0	1	4	9
$g(x) = x $						

One table has undefined entries, the other has repeats. Explain what each tells you:

$\sqrt{x}$ has a restricted _____; $|x|$ is not _____, which will matter on Day 12.

Part 2 — Absolute Value as a Piecewise Rule

Absolute value is distance from zero, so it never returns a negative. Written as two cases:

$|x| =$ _____ when $x \geq 0$, and $|x| =$ _____ when $x < 0$.

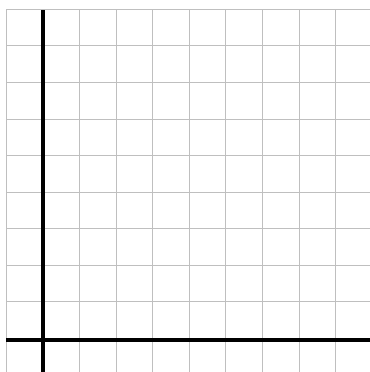
The second case looks like it produces a negative. Explain why it does not:

If x is already negative, _____. For example $-(-5) = 5$.

Part 3 — Graph Them Both

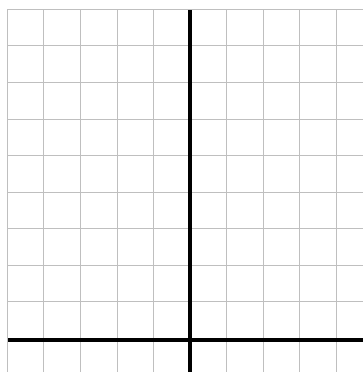
Each square is 1 unit. Plot from your tables, then connect smoothly.

A. $f(x) = \sqrt{x}$ x from 0 to 9



Domain _____ Range _____

B. $g(x) = |x|$ x from -5 to 5



Domain _____ Range _____

Part 4 — Solution Counts

Solve each. The number of answers is the point, not just the answers.

1. $|x| = 6 \rightarrow x =$ _____ how many? _____ 2. $\sqrt{x} = 6 \rightarrow x =$ _____ how many? _____

3. $|x| = 0 \rightarrow x =$ _____ how many? _____ 4. $|x| = -3 \rightarrow$ _____

Item 4 has no solution. What does that say about the range of $|x|$? _____

Connect item 1 to the graph: why do two x -values give the same output?

The V is symmetric about the _____, so a horizontal line meets it twice.

Part 5 — Going Further

1. Give the domain of each, in interval notation.

$\sqrt{x-7}$ _____ $\sqrt{3x+6}$ _____ $|x-7|$ _____

2. Why does the absolute value one have no restriction while the radicals do?

3. On the interval $[1, 9]$, give the minimum and maximum of $\sqrt{x}$. min _____ max _____

4. Is $\sqrt{x^2}$ the same function as $|x|$? Test $x = -5$ and $x = 5$, then decide.

Yes — _____

Exit Ticket

Both parent functions have minimum 0 and pass through the origin.

Name one attribute that separates them: _____ and one they share besides those two:

Write $|x|$ as a piecewise rule from memory: _____